High Level Design Document

Introduction

This High Level Design (HLD) document outlines the architecture and core components for **MedBotX - Advanced Medical Chatbot with Memory**. MedBotX is a healthcare chatbot leveraging LangChain and OpenAI to answer patient queries, maintain conversational context, and support both temporary and permanent memory across user sessions.

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1. System Architecture Overview

Architecture Description:
MedBotX is a modular, service-oriented application built with FastAPI. It integrates LangChain for conversational orchestration, OpenAI for generative responses, and memory modules for context retention. The system exposes RESTful endpoints for client interaction.

Main System Components:

Component	Role
API Layer	Handles HTTP requests/responses, authentication, and routing
Conversation Engine	Manages dialogue flow using LangChain and orchestrates memory access
Memory Module	Stores/retrieves temporary (session) and permanent (user) context
LLM Integration	Connects to OpenAI for generating responses
Data Store	Persists permanent memory and user/session data

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2. Component Interactions

Sequence Step	Interaction Description
1	Client sends query to API Layer
2	API Layer authenticates and forwards request to Conversation Engine
3	Conversation Engine retrieves context from Memory Module
4	Conversation Engine invokes LLM Integration (OpenAI) with context and user query
5	LLM Integration returns generated response to Conversation Engine
6	Conversation Engine updates Memory Module with new context
7	API Layer returns response to client

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3. Data Flow Overview

Data Flow	Source	Destination	Purpose
User Query	Client	API Layer	Initiate conversation
Context Retrieval	Memory Module	Conversation Engine	Provide session/user context
LLM Request	Conversation Engine	LLM Integration	Generate response using OpenAI
Response Delivery	LLM Integration	Conversation Engine	Return generated answer
Memory Update	Conversation Engine	Memory Module	Store new context
Response to User	API Layer	Client	Deliver answer

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4. Technology Stack

Layer/Function	Technology/Framework
API Framework	FastAPI (Python)
Conversational Flow	LangChain
LLM Integration	OpenAI API
Memory Management	LangChain Memory, Database
Data Persistence	Relational/NoSQL Database
Language	Python

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5. Scalability & Reliability

- **Scalability:**
  - Stateless API layer enables horizontal scaling.
  - Memory and data store can be distributed for high availability.
- **Reliability:**
  - Persistent storage ensures context is maintained across sessions.
  - API authentication and input validation for security.
- **Security:**
  - Secure API endpoints, encrypted data storage, and compliance with healthcare data standards (e.g., HIPAA, if required).

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